THE UNIVERSITY OF CHENAB

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SOFTWARE REQUIREMENTS SPECIFICATION

(SRS DOCUMENT)

for

3DPanoSphere(Your personalized 3D image generation app)

Version 1.0

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1 Introduction

In the modern digital era, social media platforms play a significant role in communication and content sharing. However, most existing platforms provide limited support for creating and displaying panoramic images and videos within the application. Users often need separate tools for capturing, editing, and sharing panoramic content. The proposed 3DPanoSphere mobile application aims to provide a centralized platform where users can create, share, and explore panoramic content along with standard social media features, making visual storytelling more interactive and accessible.

2 Purpose

The purpose of this project is to develop an all-in-one social media mobile application specifically designed for users interested in panoramic content creation, such as photographers, travel bloggers, influencers, and general users. The application will integrate panorama creation tools with common social media features including posting content, following users, messaging, and personalized feeds. By combining these functionalities within a single platform, the system aims to provide a simple, user-friendly, and efficient solution for creating and sharing panoramic visual content.

3 Scope

The scope of the 3DPanoSphere application includes the development of a mobile platform that allows users to capture or upload panoramic images and videos, organize them into virtual galleries, and share them with other users. The system will support social interaction features such as likes, comments, messaging, and user following, along with privacy settings and administrative controls. The application focuses on enhancing the panoramic viewing experience and building a community for users who want to explore and showcase immersive visual content.

4 Feasibility Study

4.1 Technical Feasibility Study

Technical feasibility evaluates whether the proposed 3DPanoSphere mobile application can be successfully developed using available technologies, tools, infrastructure, and technical expertise. It determines whether the system architecture, infrastructure, security mechanisms, and smart application features can be implemented within academic constraints while ensuring acceptable levels of performance, reliability, and scalability.

4.1.1 Structural Design

The structural design of the 3DPanoSphere system is based on a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. This architectural model ensures separation of concerns, modular development, scalability, and ease of maintenance.

4.1.1.1 Presentation Layer

The Presentation Layer (Mobile Application) will be developed using Flutter and will handle all user interface operations including profile management, panoramic image display, messaging interface, gallery browsing, and notifications.

4.1.1.2 Application Layer

The Application Layer (Backend Server) will be developed using Spring Boot. It will manage business logic, authentication and authorization, API communication, AI-based recommendation processing, messaging services, and database interaction.

4.1.1.3 Data Layer

The Data Layer will utilize Microsoft SQL Server for structured data storage, while high-resolution panoramic images and multimedia content will be stored in cloud storage. Only metadata references will be maintained in the relational database to improve system performance.

4.1.2 Utilities and Infrastructure

The successful implementation of 3DPanoSphere requires both hardware and software infrastructure.

4.1.2.1 Hardware Infrastructure

The project requires developer workstations for coding and testing, Android smartphones for application testing, and reliable internet connectivity for cloud integration and API communication. These resources are accessible within the academic environment.

4.1.2.2 Software Infrastructure

The system requires the following software tools and platforms:

- Flutter SDK for frontend mobile application development
- Java Development Kit (JDK) for backend development
- Spring Boot framework for RESTful API services
- Microsoft SQL Server for relational database management
- Python runtime environment for image processing and AI logic
- Cloud storage services for storing panoramic media content

All required software technologies are widely available, stable, and compatible, ensuring smooth development and deployment.

4.1.3 Safety Standards

The 3DPanoSphere application will implement security mechanisms to protect user data and prevent misuse. User passwords will be encrypted before storage, and token-based authentication will be used to secure user sessions. RESTful APIs and role-based access control will protect administrative functions, while privacy settings will allow users to control access to their profiles and galleries. Content reporting and moderation features will help maintain a safe community environment. These security measures are feasible using standard backend security frameworks and libraries.

4.1.4 Smart Application Features

The 3DPanoSphere mobile application integrates modern smart technologies to enhance user experience and system efficiency.

4.1.4.1 Panoramic Image Stitching

The system will implement panoramic image stitching algorithms that combine multiple overlapping images into a seamless panoramic image. Image alignment, feature detection, and edge blending techniques will be applied. Performance challenges such as high memory usage will be managed through image compression and resolution optimization.

4.1.4.2 AI-Based Recommendation System

A lightweight AI-based recommendation mechanism will analyze user engagement metrics such as likes, comments, views, and follows. Based on engagement scoring and content similarity, the system will generate personalized feed rankings.

4.1.4.3 Real-Time Messaging

The application will support real-time messaging between users using socket-based communication or optimized API polling methods. This enhances user interaction and platform engagement.

4.1.4.4 Virtual Gallery Management

Users will be able to create and manage virtual galleries to organize panoramic content. Relational database design with foreign keys will ensure structured linking between users and their galleries.

4.1.4.5 Analytics and Insights

The system will provide engagement analytics such as post views, likes, and follower growth trends. Aggregation queries and statistical processing will generate meaningful insights displayed in graphical format.

4.2 Financial Feasibility Study

Financial feasibility evaluates whether the 3DPanoSphere mobile application can be developed, deployed, and maintained within reasonable financial limits. It analyzes the estimated development cost, funding possibilities, projected revenue opportunities, operational expenses, and long-term sustainability. The purpose of this analysis is to determine whether the expected benefits justify the required financial investment.

4.2.1 Estimated Project Cost

The estimated initial development and deployment cost of the 3DPanoSphere application ranges from $3,000 to $5,000 for small-scale commercial deployment. This includes expenses for cloud hosting, database services, domain registration, API integration, testing devices, and initial marketing. Since the project uses open-source technologies such as Flutter and Spring Boot, software licensing costs are minimal.

However, cloud storage for high-resolution panoramic images may represent the main technical expense. For academic development, the financial cost remains very low and mostly limited to internet usage and optional cloud testing services.

4.2.2 Funding Sources

Since the project is currently developed in an academic environment, it is primarily self-financed by the development team. No large capital investment is required at this stage. If the application is expanded into a commercial product, potential funding sources may include private investors, startup incubators, government innovation grants, and technology entrepreneurship programs. These funding sources can support infrastructure scaling, marketing campaigns, and long-term operational management.

4.2.3 Projected Revenue Potential

Although currently an academic project, 3DPanoSphere has strong commercial potential due to its focus on panoramic digital storytelling. Revenue could be generated through premium subscriptions, advertisement integration, featured gallery promotions, and collaborations with professional photographers. If the platform gains sufficient user engagement, it may generate an estimated annual revenue of $8,000 to $20,000 during its early commercial stage, with higher scalability as the user base grows.

4.2.4 Break-even Analysis

Due to relatively low initial investment and moderate operational expenses, the break-even point could be achieved within 2 to 3 years of commercial launch, depending on user adoption and monetization success. The digital structure of the system reduces fixed costs such as physical infrastructure and logistics, which improves profitability potential compared to traditional businesses.

4.2.5 Operational Expenses

The primary ongoing operational expenses include:

- Cloud hosting and storage services
- Server maintenance and security monitoring
- System upgrades and technical support
- Marketing and promotional activities

Among these, cloud storage remains the most significant recurring cost because panoramic images require high storage capacity. However, storage plans can be scaled according to user growth, allowing cost control and financial flexibility.

4.3 Operational Feasibility Study

Operational feasibility evaluates whether the 3DPanoSphere application can function effectively in a real-world environment. The system is designed to support user interaction, administrative control, and efficient platform management. By providing familiar social media features along with panoramic content sharing, the application is expected to be easily accepted by its target users. The analysis indicates that the system can be operated and maintained using available technical resources.

4.3.1 Application Management System

The system will include an integrated administrative management structure to monitor platform activities and maintain community guidelines. Administrators will be able to manage user accounts, review reported content, and take corrective actions such as removing inappropriate posts or suspending accounts. Activity logs and monitoring tools will support transparency and effective governance, ensuring smooth operation of the platform.

4.3.2 Operational Resources and Infrastructure

The system relies on stable cloud hosting services, database management systems, and secure internet connectivity to operate efficiently. Since the application is cloud-based, it does not require physical infrastructure beyond server hosting and storage services. The required operational resources include:

- Cloud storage for panoramic media
- Backend hosting services
- Database server management
- Network connectivity for API communication

These resources are widely available and scalable, ensuring that operational demands can be adjusted according to user growth.

4.3.3 Staffing and Maintenance Plan

For academic deployment, the development team will handle system monitoring and maintenance. If the system is commercialized, a small technical team will manage backend monitoring, security updates, bug fixes, and system improvements. The modular architecture of the application allows separate maintenance of frontend, backend, and database components, reducing operational complexity and making the system easier to maintain and upgrade.

4.3.4 User Acceptance

User acceptance is expected to be high because the target users—photographers, travel bloggers, content creators, and general social media users—are already familiar with common social media features such as posting, liking, commenting, following, and messaging. 3DPanoSphere follows a similar interaction model while introducing specialized panoramic content features and virtual galleries, making the transition easy for users and reducing the learning curve.

4.3.5 Ease of Use and Usability

The application is designed with a simple and intuitive interface to ensure ease of use. Users will be able to easily navigate profiles, galleries, messages, and content feeds. The panoramic creation module will guide users through the process of capturing and uploading images, while privacy settings will be managed through simple options. These design considerations ensure that both technical and non-technical users can comfortably operate the system.

4.3.6 User Support Strategy

To ensure smooth operation, the application will include built-in support features such as content reporting, feedback submission, and automated notifications. Users will be able to report inappropriate content or technical issues directly through the app. These mechanisms will help maintain platform quality and ensure timely issue resolution. In a commercial deployment, additional support options such as email assistance or chatbot services can further enhance user experience.

4.4 Schedule Feasibility Study

Schedule feasibility evaluates whether the 3DPanoSphere mobile application can be developed, tested, and deployed within the planned academic or operational timeline. It ensures that project milestones are realistic, achievable, and aligned with resource availability. This assessment also identifies potential risks that may delay completion and proposes strategies to mitigate them.

4.4.1 Planning and Design Phase

The initial phase involves detailed requirements gathering, system architecture design, and user interface planning. This includes:

- Identifying functional and non-functional requirements

- Creating UML diagrams (Use Case, Class, Sequence, Activity)
- Designing wireframes for the mobile application

The planning and design phase is expected to take **1-2 months**, allowing sufficient time to validate project scope and ensure that the architecture aligns with technical, financial, and operational feasibility.

4.4.2 Development Phase

The development phase includes the implementation of the frontend using Flutter, backend using Spring Boot, database setup with SQL Server, and AI-based recommendation module development.

Key activities during this phase:

- User authentication and registration module
- Panoramic image creation and gallery module
- Real-time messaging system
- Analytics and insights module

This phase is estimated to take **3-4 months**, accounting for integration of all modules, code reviews, and incremental testing during development.

4.4.3 Testing and Quality Assurance

After development, a dedicated **testing period of 1 month** will be allocated for:

- Unit testing of individual modules
- Integration testing to ensure seamless communication between frontend, backend, and database
- User acceptance testing to validate usability and feature functionality

This ensures that the final application is stable, secure, and meets project requirements.

4.4.4 Deployment and User Training

Deployment involves launching the application on the target mobile platforms (Android/iOS) and setting up cloud infrastructure for data storage and processing. User support resources, guided tutorials, and documentation will be provided to ensure smooth onboarding. This phase is planned for 1 month, including minor post-deployment updates and bug fixes.

4.4.5 Project Completion Timeline

The overall project is expected to be completed within **6-8 months** from the start of the **planning phase**. This schedule is realistic given the modular architecture, availability of development tools, and the academic resources accessible to the project team.

- Risk mitigation strategies for potential delays include:

- Parallel development of frontend and backend components
- Early integration of AI and image processing modules
- Regular milestone reviews to identify and resolve bottlenecks promptly

4.5 Legal Feasibility Study

Legal feasibility evaluates whether the proposed 3DPanoSphere mobile application complies with relevant laws, digital policies, and online content regulations. It ensures that the system operates legally with respect to data privacy, intellectual property rights, user responsibility, and messaging compliance. This assessment helps mitigate potential legal risks during both academic deployment and possible commercialization.

4.5.1 Data Privacy and Protection

The 3DPanoSphere application collects user information such as email addresses, usernames, passwords, and multimedia content, making data privacy a key legal concern. To protect user accounts, the system will implement password encryption and secure authentication mechanisms. Users will also have privacy settings that allow them to control the visibility of their profiles and galleries with public or private options.

Additionally, user data will not be shared with third parties without consent. These measures follow standard data protection principles such as confidentiality, integrity, and controlled access, reducing the risk of data breaches.

4.5.2 Intellectual Property and Content Ownership

3DPanoSphere allows users to upload panoramic images, videos, and other digital content while maintaining their ownership rights over the media they create. The platform's Terms and Conditions will clearly state that the system only hosts the content and does not claim ownership. Users will be responsible for ensuring that they have the legal rights to upload any media. Uploading copyrighted material without authorization will be prohibited to protect both the platform and its users from intellectual property violations.

4.5.3 Content Regulation and Community Guidelines

To prevent misuse and legal complications, 3DPanoSphere will establish clear community guidelines. These guidelines prohibit:

- Offensive, abusive, or harmful content
- Harassment, cyberbullying, or threats
- Misleading, illegal, or copyright-infringing material

A reporting system will enable users to flag inappropriate content, and the administrative panel will allow moderators to review and take action, including content removal or account suspension. These measures ensure compliance with digital content laws and maintain a safe platform environment.

4.5.4 Digital Communication Compliance

Messaging and communication features must comply with general online regulations. The system will prevent illegal activities such as fraud, threats, or abusive communications. User safety is reinforced through reporting and blocking features. Additionally, basic age restrictions may be applied, such as requiring a minimum user age, to ensure compliance with child protection standards and other relevant legal frameworks.

4.5.5 Legal Documentation

To provide transparency and legal clarity, the platform will include the following legal documents:

Privacy Policy: Details how user data is collected, stored, and used

Terms and Conditions: Defines user responsibilities, rights, and obligations

Community Guidelines: Specifies acceptable behavior and content rules

These documents reduce legal ambiguity, ensure compliance with digital policies, and protect both users and the platform from potential disputes.

4.6 Social Feasibility Study

Social feasibility evaluates the impact of the 3DPanoSphere mobile application on its users, the broader community, and society at large. It examines whether the system is socially acceptable, whether it encourages positive behaviors such as creativity and collaboration, and whether any negative consequences may arise. This analysis ensures that the platform provides value to users while minimizing risks to both individual well-being and the online community.

4.6.1 Positive Social Impact

The 3DPanoSphere application encourages creative expression by providing a platform where users can create, organize, and share panoramic visual content. It supports photographers, travel bloggers, content creators, and hobbyists by allowing them to showcase their work through immersive visual storytelling. The virtual gallery feature enables users to present their content in an organized and visually appealing way, promoting knowledge sharing and creativity. Additionally, social features such as messaging, following, notifications, and content engagement help build a collaborative community and increase networking opportunities among users.

4.6.2 Potential Negative Social Impact

Despite its advantages, the platform may also present certain social challenges such as excessive screen time, cyberbullying, harassment, or the misuse of copyrighted material. Inappropriate content may also appear if proper moderation is not maintained. To reduce these risks, 3DPanoSphere will implement community guidelines, content reporting systems, privacy controls, and administrator moderation.

4.6.3 Social Acceptance and Usability

The platform is designed with familiar social media interaction patterns, similar to Instagram or Facebook, which encourages high user acceptance. Features like guided panorama creation, intuitive navigation, and customizable privacy settings make the application approachable for users with varying technical skills. By combining a familiar interface with innovative content features, 3DPanoSphere ensures both usability and social acceptability, promoting consistent engagement across user demographics.

4.6.4 Community and Environmental Considerations

While primarily a digital platform, social feasibility also considers indirect community and environmental effects:

Sustainability Practices: Use of energy-efficient hosting and cloud services to reduce digital carbon footprint.

Community Engagement: Encourages collaboration between users, photographers, and small creative businesses.

Content Awareness: Promotes positive storytelling and respectful interaction, reducing exposure to harmful digital behavior.

5 Functional Requirements

5.1 User Class 1 - New User

5.1.1.1 Functional Requirement 1.1

ID: NU-FE-1

TITLE: Explore app without registration

DESC: New or guest users can open the 3DPanoSphere app and browse publicly available panoramic content and virtual galleries without creating an account.

RAT: To allow potential users to explore the app features and understand its value before registering.

DEP: Requires access to public galleries and content databases.

5.1.1.2 Functional Requirement 1.2

ID: NU-FE-2

TITLE: Content viewing

DESC: Guest users can view public panoramic images, videos, and galleries, but cannot interact (like, comment, or share).

RAT: Encourages app engagement and showcases platform's panoramic capabilities to attract new users.

DEP: Public content must be accessible and privacy settings respected.

5.1.1.3 Functional Requirement 1.3

ID: NU-FE-3

TITLE: Search users and galleries

DESC: Guests can search for registered users or public virtual galleries using keywords or hashtags, enabling them to explore content of interest.

RAT: Helps new users find content and creators that align with their interests, improving engagement.

DEP: Requires indexed database of users and galleries with search functionality.

5.1.1.4 Functional Requirement 1.4

ID: NU-FE-4

TITLE: Limited story viewing

DESC: Guests can view public stories and time-limited content, but cannot react, comment, or create stories themselves.

RAT: Introduces users to dynamic content features and panoramic storytelling.

DEP: Requires story module with public access permissions.

5.1.1.5 Functional Requirement 1.5

ID: NU-FE-5

TITLE: Registration prompt

DESC: When guests try to interact (like, comment, follow, or message), the system will prompt them to register or log in to unlock full functionality.

RAT: Encourages conversion of new users into registered users, increasing engagement and platform growth.

DEP: Requires registration module and user authentication system.

5.1.1.6 Functional Requirement 1.6

ID: NU-FE-6

TITLE: Account registration access

DESC: Guests can initiate account creation through email, phone number, or social media sign-in options.

RAT: Provides a smooth onboarding experience to transition from guest to registered user.

DEP: Requires functional registration and verification modules.

5.1.1.7 Functional Requirement 1.7

ID: NU-FE-7

TITLE: App tour / onboarding

DESC: New users can access an interactive app tour or tutorial showing key features like panorama creation, galleries, and stories.

RAT: Reduces learning curve and ensures users understand app benefits before full interaction.

DEP: Requires front-end UI walkthrough and instructional content.

5.1.1.8 Functional Requirement 1.8

ID: NU-FE-8

TITLE: Privacy awareness

DESC: Display privacy policy, terms and conditions, and content ownership rules during initial guest interaction or first registration prompt.

RAT: Ensures legal compliance and informs new users about platform rules before registration.

DEP: Requires properly formatted legal documents accessible via UI.

5.2 User Class 2 - Registered User

5.2.1.1 Functional Requirement 2.1

ID: FR-RU-01

TITLE: User registration

DESC: Allow users to register using email, phone number, or social media accounts.

RAT: Provides secure entry to the platform.

DEP: User management module, authentication system

5.2.1.2 Functional Requirement 2.2

ID: FR-RU-02

TITLE: User login

DESC: Allow users to log in using registered credentials with two-factor authentication.

RAT: Ensures secure access and identity verification.

DEP: Authentication module

5.2.1.3 Functional Requirement 2.3

ID: FR-RU-03

TITLE: Password reset

DESC: Users can reset forgotten passwords via email or phone verification.

RAT: Supports account recovery in case of lost credentials.

DEP: User management module, email/SMS service

5.2.1.4 Functional Requirement 2.4

ID: FR-RU-04

TITLE: Logout from all devices

DESC: Users can log out remotely from all active sessions.

RAT: Enhances account security.

DEP: Session management module

5.2.1.5 Functional Requirement 2.5

ID: FR-RU-05

TITLE: Account deactivation/deletion

DESC: Users can deactivate or permanently delete their accounts.

RAT: Gives control over account lifecycle and privacy.

DEP: User management module

5.2.1.6 Functional Requirement 2.6

ID: FR-RU-06

TITLE: Profile creation

DESC: Users can create and fill personal information like name, bio, location, and contact info.

RAT: Personalizes user experience.

DEP: Profile database

5.2.1.7 Functional Requirement 2.7

ID: FR-RU-07

TITLE: Profile picture upload

DESC: Users can upload or update profile and cover images.

RAT: Visual identity for social interaction.

DEP: Media module

5.2.1.8 Functional Requirement 2.8

ID: FR-RU-08

TITLE: Privacy settings

DESC: Users can set account privacy (public/private) and manage post visibility.

RAT: Protects user content and personal data.

DEP: Privacy management module

5.2.1.9 Functional Requirement 2.9

ID: FR-RU-09

TITLE: Notification preferences

DESC: Users can configure alerts for likes, comments, messages, and system notifications.

RAT: Enables personalized communication.

DEP: Notification module

5.2.1.10 Functional Requirement 2.10

ID: FR-RU-10

TITLE: Follow/unfollow users

DESC: Users can follow or unfollow other users to manage their feed.

RAT: Supports content discovery and social interaction.

DEP: User relationship module

5.2.1.11 Functional Requirement 2.11

ID: FR-RU-11

TITLE: Block/unblock users

DESC: Users can block or unblock accounts to control interactions.

RAT: Ensures a safe and comfortable environment.

DEP: User relationship module

5.2.1.12 Functional Requirement 2.12

ID: FR-RU-12

TITLE: Capture panorama

DESC: Users can capture multiple overlapping images for panoramic content.

RAT: Core feature for immersive visual storytelling.

DEP: Camera and media processing module

5.2.1.13 Functional Requirement 2.13

ID: FR-RU-13

TITLE: Auto-stitching

DESC: Images are automatically aligned and stitched into a panoramic image.

RAT: Simplifies panorama creation for users.

DEP: Image processing module

5.2.1.14 Functional Requirement 2.14

ID: FR-RU-14

TITLE: Panorama preview and save

DESC: Users can preview and save panoramic images to the gallery or device.

RAT: Allows users to review content before sharing.

DEP: Storage module

5.2.1.15 Functional Requirement 2.15

ID: FR-RU-15

TITLE: Panorama sharing

DESC: Users can share panoramic images directly to their profile or external platforms.

RAT: Supports social engagement and visibility.

DEP: Media sharing module

5.2.1.16 Functional Requirement 2.16

ID: FR-RU-16

TITLE: Upload image/video

DESC: Users can upload images or videos to their profile or galleries.

RAT: Allows content sharing and social engagement.

DEP: Media upload module

5.2.1.17 Functional Requirement 2.17

ID: FR-RU-17

TITLE: Apply filters & effects

DESC: Users can apply visual filters, effects, and edit uploaded media.

RAT: Enhances visual appeal and creativity.

DEP: Media editing module

5.2.1.18 Functional Requirement 2.18

ID: FR-RU-18

TITLE: Add captions & tags

DESC: Users can add descriptive captions, hashtags, or tags to their posts.

RAT: Improves content discoverability and context.

DEP: Content management module

5.2.1.19 Functional Requirement 2.19

ID: FR-RU-19

TITLE: Post visibility settings

DESC: Users can select whether a post is public, private, or visible to specific users.

RAT: Maintains user control over content audience.

DEP: Privacy module

5.2.1.20 Functional Requirement 2.20

ID: FR-RU-20

TITLE: Delete posts

DESC: Users can remove any post from their profile or gallery.

RAT: Ensures content management flexibility.

DEP: Content management module

5.2.1.21 Functional Requirement 2.21

ID: FR-RU-21

TITLE: Create time-limited stories

DESC: Users can create stories that expire after a set duration.

RAT: Encourages frequent engagement and content sharing.

DEP: Story module

5.2.1.22 Functional Requirement 2.22

ID: FR-RU-22

TITLE: Story customization

DESC: Add text, stickers, filters, or panoramic content to stories.

RAT: Enhances creativity and immersive experience.

DEP: Story editing module

5.2.1.23 Functional Requirement 2.23

ID: FR-RU-23

TITLE: Story highlights

DESC: Users can save selected stories permanently in highlights.

RAT: Allows long-term showcase of important content.

DEP: Story storage module

5.2.1.24 Functional Requirement 2.24

ID: FR-RU-24

TITLE: Story analytics

DESC: Users can view viewer count and interactions on their stories.

RAT: Provides insight into engagement and reach.

DEP: Analytics module

5.2.1.25 Functional Requirement 2.25

ID: FR-RU-25

TITLE: Delete story

DESC: Users can remove a story at any time before it expires.

RAT: Provides control over content lifecycle.

DEP: Story module

5.2.1.26 Functional Requirement 2.26

ID: FR-RU-26

TITLE: Create virtual gallery

DESC: Users can create dedicated gallery spaces to showcase their panoramic content.

RAT: Provides a structured platform for immersive visual storytelling.

DEP: Virtual exhibition module

5.2.1.27 Functional Requirement 2.27

ID: FR-RU-27

TITLE: Customize gallery layouts & themes

DESC: Users can personalize gallery design, layout, and theme.

RAT: Enhances user experience and branding.

DEP: Virtual exhibition module

5.2.1.28 Functional Requirement 2.28

ID: FR-RU-28

TITLE: Organize & curate content

DESC: Users can sort, categorize, and arrange their uploaded content.

RAT: Improves content discoverability and user experience.

DEP: Gallery management module

5.2.1.29 Functional Requirement 2.29

ID: FR-RU-29

TITLE: Visitor analytics

DESC: Users can track visitor counts and interactions with their gallery.

RAT: Provides feedback on content popularity.

DEP: Analytics module

5.2.1.30 Functional Requirement 2.30

ID: FR-RU-30

TITLE: Send direct messages

DESC: Users can send text, audio, and video messages to other users.

RAT: Enables private communication and collaboration.

DEP: Messaging module

5.2.1.31 Functional Requirement 2.31

ID: FR-RU-31

TITLE: Create groups

DESC: Users can create groups for multiple users to communicate together.

RAT: Supports collaborative interactions and community building.

DEP: Messaging module

5.2.1.32 Functional Requirement 2.32

ID: FR-RU-32

TITLE: Add/remove members

DESC: Users can add or remove members from their groups.

RAT: Manages group participation and control.

DEP: Group management module

5.2.1.33 Functional Requirement 2.33

ID: FR-RU-33

TITLE: Pin/delete chats

DESC: Users can pin important chats or delete them from their inbox.

RAT: Enhances usability and chat management.

DEP: Messaging module

5.2.1.34 Functional Requirement 2.34

ID: FR-RU-34

TITLE: View post engagement metrics

DESC: Users can view likes, comments, and shares for their posts.

RAT: Provides insight into how well their content is performing.

DEP: Analytics module

5.2.1.35 Functional Requirement 2.35

ID: FR-RU-35

TITLE: Content performance analysis

DESC: Users can see detailed performance trends of their posts over time.

RAT: Helps users understand audience preferences and optimize content.

DEP: Analytics module

5.2.1.36 Functional Requirement 2.36

ID: FR-RU-36

TITLE: Best posting time recommendations

DESC: System suggests optimal posting times based on audience engagement patterns.

RAT: Improves content visibility and user interaction.

DEP: Analytics module

5.2.1.37 Functional Requirement 2.37

ID: FR-RU-37

TITLE: Profile visit analytics

DESC: Users can see the number of visitors to their profile.

RAT: Measures audience reach and popularity.

DEP: Analytics module

5.2.1.38 Functional Requirement 2.38

ID: FR-RU-38

TITLE: Content reach statistics

DESC: Users can monitor how many people have viewed individual posts.

RAT: Provides insight into post engagement and visibility.

DEP: Analytics module

5.2.1.39 Functional Requirement 2.39

ID: FR-RU-39

TITLE: Audience growth tracking

DESC: Users can track the increase in followers over time.

RAT: Helps evaluate growth trends and social influence.

DEP: Analytics module

5.2.1.40 Functional Requirement 2.40

ID: FR-RU-40

TITLE: Push notifications for interactions

DESC: Users receive notifications for likes, comments, follows, and mentions.

RAT: Keeps users engaged and informed.

DEP: Notification module

5.2.1.41 Functional Requirement 2.41

ID: FR-RU-41

TITLE: Custom notification preferences

DESC: Users can enable or disable notifications for specific events.

RAT: Provides user control over notification preferences.

DEP: Notification module

5.2.1.42 Functional Requirement 2.42

ID: FR-RU-42

TITLE: Real-time engagement alerts

DESC: Users get instant alerts for important interactions on their content.

RAT: Enhances user responsiveness and engagement.

DEP: Notification module

5.2.1.43 Functional Requirement 2.43

ID: FR-RU-43

TITLE: Important updates & announcements

DESC: Users are informed of system updates, policy changes, or announcements.

RAT: Keeps the community informed and compliant with platform policies.

DEP: Notification module

5.2.1.44 Functional Requirement 2.44

ID: FR-RU-44

TITLE: Account privacy controls

DESC: Users can set account visibility to private or public.

RAT: Ensures user control over who can access their content.

DEP: Privacy module

5.2.1.45 Functional Requirement 2.45

ID: FR-RU-45

TITLE: Content visibility settings

DESC: Users can manage who can see individual posts.

RAT: Enhances content privacy and security.

DEP: Privacy module

5.2.1.46 Functional Requirement 2.46

ID: FR-RU-46

TITLE: Data sharing preferences

DESC: Users can control sharing of personal data with third-party apps.

RAT: Protects user data and ensures compliance with privacy policies.

DEP: Privacy module

5.2.1.47 Functional Requirement 2.47

ID: FR-RU-47

TITLE: Account activity monitoring

DESC: Users can view login history and account activity.

RAT: Increases transparency and security awareness.

DEP: Privacy & security module

5.2.1.48 Functional Requirement 2.48

ID: FR-RU-48

TITLE: Change passwords & linked emails/phones

DESC: Users can update login credentials and contact information.

RAT: Maintains account security and user control.

DEP: Account management module

5.2.1.49 Functional Requirement 2.49

ID: FR-RU-49

TITLE: Report inappropriate content

DESC: Users can report offensive or inappropriate posts and comments.

RAT: Helps maintain community safety and standards.

DEP: Moderation module

5.2.1.50 Functional Requirement 2.50

ID: FR-RU-50

TITLE: Appeal moderation decisions

DESC: Users can appeal if their content is flagged or removed.

RAT: Ensures fairness and accountability in content moderation.

DEP: Moderation module

5.2.1.51 Functional Requirement 2.51

ID: FR-RU-51

TITLE: View community guidelines

DESC: Users can access and read platform rules and guidelines.

RAT: Promotes awareness of acceptable behavior and content standards.

DEP: Moderation module

5.3 User Class 3 - Admin User

5.3.1.1 Functional Requirement 3.1

ID: FR-AU-1

TITLE: View all registered users

DESC: Admin can view a complete list of all registered users, including profile details and account status.

RAT: Enables monitoring and management of the platform's user base.

DEP: User management module

5.3.1.2 Functional Requirement 3.2

ID: FR-A-2U

TITLE: Block/unblock users

DESC: Admin can block users violating platform rules or unblock previously blocked users.

RAT: Ensures compliance with community guidelines and platform safety.

DEP: User management module

5.3.1.3 Functional Requirement 3.3

ID: FR-AU-3

TITLE: Delete user accounts

DESC: Admin can permanently remove user accounts that are inactive or violate rules.

RAT: Maintains platform integrity and reduces spam.

DEP: User management module

5.3.1.4 Functional Requirement 3.4

ID: FR-AU-4

TITLE: Verify user accounts

DESC: Admin can approve verification requests for users (email/phone/social media verification).

RAT: Confirms authenticity of user accounts and reduces fake profiles.

DEP: User management module

5.3.1.5 Functional Requirement 3.5

ID: FR-AU-5

TITLE: Review reported content

DESC: Admin can access all content flagged by users and review it for compliance.

RAT: Ensures timely removal of inappropriate material.

DEP: Moderation module

5.3.1.6 Functional Requirement 3.6

ID: FR-AU-6

TITLE: Warn or block users

DESC: Admin can issue warnings or block users based on repeated content violations.

RAT: Maintains community standards and deters misconduct.

DEP: Moderation module

5.3.1.7 Functional Requirement 3.7

ID: FR-AU-7

TITLE: Enforce community guidelines

DESC: Admin can update and enforce rules regarding acceptable content and behavior.

RAT: Keeps the platform safe and socially responsible.

DEP: Moderation module

5.3.1.8 Functional Requirement 3.8

ID: FR-AU-8

TITLE: Manage appeals

DESC: Admin can review and act on user appeals regarding content removal or account actions.

RAT: Ensures fairness and transparency in moderation.

DEP: Moderation module

5.3.1.9 Functional Requirement 3.9

ID: FR-AU-9

TITLE: Platform usage analytics

DESC: Admin can view overall platform usage metrics such as daily active users, post engagement, and virtual gallery visits.

RAT: Helps in monitoring platform performance and growth trends.

DEP: Analytics module

5.3.1.10 Functional Requirement 3.10

ID: FR-AU-10

TITLE: Content engagement reports

DESC: Admin can see which posts, galleries, or stories are most engaged with by users.

RAT: Identifies trending content and informs moderation or feature improvements.

DEP: Analytics module

5.3.1.11 Functional Requirement 3.11

ID: FR-AU-11

TITLE: User activity reports

DESC: Admin can access reports of individual user activity, including login history and post interactions.

RAT: Helps detect suspicious or abusive behavior.

DEP: Analytics module

5.3.1.12 Functional Requirement 3.12

ID: FR-AU-12

TITLE: Update system configurations

DESC: Admin can modify platform settings such as notification preferences, security configurations, and feature toggles.

RAT: Ensures smooth operation and adaptability of the system.

DEP: System settings module

5.3.1.13 Functional Requirement 3.13

ID: FR-AU-13

TITLE: Manage content templates and filters

DESC: Admin can add or modify filters, categories, or templates for posts and virtual galleries.

RAT: Keeps content presentation consistent and user-friendly.

DEP: System settings module

5.3.1.14 Functional Requirement 3.14

ID: FR-AU-14

TITLE: System backup and restore

DESC: Admin can perform periodic backups and restore data in case of system failures.

RAT: Ensures data integrity and disaster recovery.

DEP: System settings module

5.3.1.15 Functional Requirement 3.15

ID: FR-AU-15

TITLE: Broadcast important updates

DESC: Admin can send platform-wide notifications, including announcements, maintenance alerts, or policy changes.

RAT: Keeps users informed and engaged with platform changes.

DEP: Notification module

5.3.1.16 Functional Requirement 3.16

ID: FR-AU-16

TITLE: Custom alerts for critical events

DESC: Admin receives notifications for urgent events such as security breaches, system errors, or mass reports of inappropriate content.

RAT: Enables rapid response and platform protection.

DEP: Notification module

6 Non-Functional Requirements

6.1 Performance

ID: QR1

TAG: System Response Time

GIST: Response time of the system when users interact with panoramic content and galleries.

SCALE: The time required for loading panoramic images, user profiles, and virtual galleries.

METER: Measurements obtained from 1000 user requests during system testing.

MUST: The system should respond within 3 seconds.

ID: QR2

TAG: Media Loading Performance

GIST: Performance of loading high-resolution panoramic images and media.

SCALE: The time required to load 360-degree panoramic images and videos.

METER: Measurements obtained from 500 media loading tests during testing.

MUST: Media content should load within 5 seconds.

ID: QR3

TAG: Concurrent User Performance

GIST: System performance when multiple users access the platform simultaneously.

SCALE: Number of users that can use the system without performance degradation.

METER: Load testing with simulated concurrent users.

MUST: The system should support 500 concurrent users.

ID: QR4

TAG: Data Processing Performance

GIST: Performance of processing user actions such as uploading panoramas, liking, commenting, and searching.

SCALE: Time required for processing user requests.

METER: Measurements obtained from 1000 system operations during testing.

MUST: Operations should be processed within 3 seconds.

ID: QR5

TAG: Search Performance

GIST: Performance of the search feature when users search for galleries, users, or panoramic content.

SCALE: Time required for search results to appear.

METER: Measurements obtained from 1000 search queries during testing.

MUST: Search results should appear within 3 seconds.

6.2 Usability

ID: QR6

TAG: System Usability

GIST: Ease of use of the 3DPanoSphere application for new and registered users.

SCALE: The ability of users to navigate the system, view panoramic galleries, and access features without confusion.

METER: Usability testing performed with 100 test users performing common tasks such as browsing galleries and viewing panoramas.

MUST: At least 85% of users should complete tasks successfully without assistance.

ID: QR7

TAG: User Interface Clarity

GIST: Clarity and simplicity of the user interface.

SCALE: The interface should present clear navigation menus, icons, and instructions.

METER: Usability testing with 100 users evaluating interface clarity.

MUST: At least 85% of users should report the interface as easy to understand.

ID: QR8

TAG: Navigation Efficiency

GIST: Efficiency of navigation within the application.

SCALE: Users should be able to reach major features such as gallery viewing, profile pages, and search within a small number of steps.

METER: Task completion testing with 100 users during usability evaluation.

MUST: Users should reach major features within 3 navigation steps.

ID: QR9

TAG: User Learning Time

GIST: Time required for new users to understand and start using the system.

SCALE: The amount of time required for a new user to learn basic functions such as viewing panoramas and browsing galleries.

METER: Usability testing with new users during system trials.

MUST: New users should understand basic system functions within 10 minutes.

ID: QR10

TAG: Accessibility Support

GIST: Accessibility support for different users.

SCALE: The application should support accessibility features such as readable text, clear icons, and responsive layouts.

METER: Accessibility evaluation based on standard usability guidelines.

MUST: The interface should support readable fonts and responsive design for different screen sizes.

6.3 Reliability

ID: QR11

TAG: System Reliability

GIST: The reliability of the 3DPanoSphere system during normal operation.

SCALE: The system should correctly display panoramic images, user profiles, and galleries without system failure.

METER: Measurements obtained from 1000 system operations during testing.

MUST: More than 97% of operations should produce correct results without errors.

ID: QR12

TAG: System Error Handling

GIST: The system's ability to handle errors and unexpected situations.

SCALE: The system should detect errors and display proper error messages without crashing.

METER: Testing with simulated system errors during 500 system operations.

MUST: The system should display proper error messages in 100% of detected errors.

ID: QR13

TAG: Data Consistency

GIST: Consistency of user data and panoramic media in the system.

SCALE: User data such as profile information, uploaded panoramas, and gallery content should remain accurate and consistent across sessions.

METER: Measurements obtained from 1000 data update operations during testing.

MUST: Data consistency should be maintained in 98% of operations.

ID: QR14

TAG: System Recovery

GIST: Ability of the system to recover after system failure or crash.

SCALE: The system should restore services and recover stored data after failure.

METER: Recovery tests performed after simulated system failures during testing.

MUST: System services should be restored within 5 minutes after failure.

6.4 Security

ID: QR15

TAG: Communication Security

GIST: Security of communication between the user device and the system server.

SCALE: All login, registration, and data communication between the client and server should be encrypted.

METER: Security testing performed on 1000 login sessions during system testing.

MUST: 100% of login communication messages must be encrypted.

ID: QR16

TAG: User Account Security

GIST: Security of registered user accounts.

SCALE: If a user attempts to log in with incorrect credentials, the system should deny access and notify the user.

METER: 1000 login attempts with invalid credentials during testing.

MUST: 100% of invalid login attempts should be rejected by the system.

ID: QR17

TAG: Admin Account Security

GIST: Security of admin accounts.

SCALE: If an admin attempts to log in with incorrect credentials, the system must prevent login and display a login failure message.

METER: 1000 invalid admin login attempts during testing.

MUST: 100% of invalid login attempts must be blocked.

ID: QR18

TAG: Account Lock Security

GIST: Protection against repeated failed login attempts.

SCALE: If a user or admin enters incorrect login credentials three consecutive times, the account should be temporarily locked.

METER: 1000 login attempts after account lock during testing.

MUST: The account should be locked for 30 minutes after three failed login attempts.

ID: QR19

TAG: Data Privacy Security

GIST: Protection of user personal information stored in the system.

SCALE: Personal data such as email, password, and profile information must be securely stored and protected.

METER: Security testing of 1000 stored user records during testing.

MUST: 100% of sensitive user data must be securely stored using encryption.

ID: QR20

TAG: Content Reporting Security

GIST: Protection of users from inappropriate or harmful content.

SCALE: Users should be able to report inappropriate content to the system administrators.

METER: Testing of 500 content reporting actions during system testing.

6.5 Scalability

ID: QR21

TAG: System Scalability

GIST: The ability of the system to handle increasing numbers of users and content.

SCALE: The system should maintain stable performance when the number of users and panoramic media increases.

METER: Load testing with simulated concurrent users during system testing.

MUST: The system should support at least 500 concurrent users without performance degradation.

ID: QR22

TAG: Media Storage Scalability

GIST: Scalability of storage for panoramic images and multimedia content.

SCALE: The system should store and manage increasing volumes of panoramic media without affecting performance.

METER: Storage testing with large volumes of panoramic images during testing.

MUST: The system should support storage of at least 10,000 panoramic media files.

ID: QR23

TAG: Database Scalability

GIST: Scalability of the database when the number of users and data grows.

SCALE: The database should support increasing numbers of users, galleries, and media content.

METER: Database load testing with simulated user and content growth.

MUST: The database should support 10,000 registered users without performance degradation.

6.6 Maintainability

ID: QR24

TITLE: Application maintainability

DESC: The application should be designed using modular architecture so developers can easily modify, update, or extend system components.

RAT: To allow efficient maintenance and future system upgrades.

DEP: None

ID: QR25

TITLE: Application testability

DESC: Testing environments and debugging tools should be available for testing system functions such as login, gallery viewing, and media upload.

RAT: To ensure proper testing and error detection during development and maintenance.

DEP: None

ID: QR26

TITLE: Code documentation

DESC: The system source code should include proper documentation and comments to support future development and maintenance.

RAT: To help developers understand the system structure and perform modifications easily.

DEP: None

6.7 Compliance

ID: QR27

TITLE: Data privacy compliance

DESC: The system must protect user personal information such as email, profile details, and uploaded panoramic media according to standard data privacy regulations.

RAT: To ensure protection of user data and compliance with privacy policies.

DEP: Privacy policy and data protection mechanisms.

ID: QR28

TITLE: Intellectual property compliance

DESC: Users retain ownership of the panoramic images, galleries, and other media they upload to the platform.

RAT: To comply with intellectual property laws and protect user content rights.

DEP: Terms and conditions module.

ID: QR29

TITLE: Terms and conditions agreement

DESC: Users must accept the platform's terms and conditions before creating an account or uploading content.

RAT: To ensure users understand the platform policies and responsibilities.

DEP: User registration module.

6.8 Social & Ethical

ID: QR30

TITLE: Community guidelines enforcement

DESC: The system should provide mechanisms for reporting inappropriate, offensive, or harmful content posted by users.

RAT: To maintain a safe and respectful community environment.

DEP: Content moderation system and admin panel.

ID: QR31

TITLE: Prevention of harmful content

DESC: The platform should discourage posting of harmful, abusive, or misleading panoramic media and provide tools for moderation.

RAT: To ensure responsible and ethical use of the platform.

DEP: Moderation and reporting tools.

ID: QR32

TITLE: User behavior monitoring

DESC: The system should monitor user activities to detect repeated violations of community guidelines.

RAT: To reduce harassment, spam, and misuse of the platform.

DEP: Moderation system and activity logs.